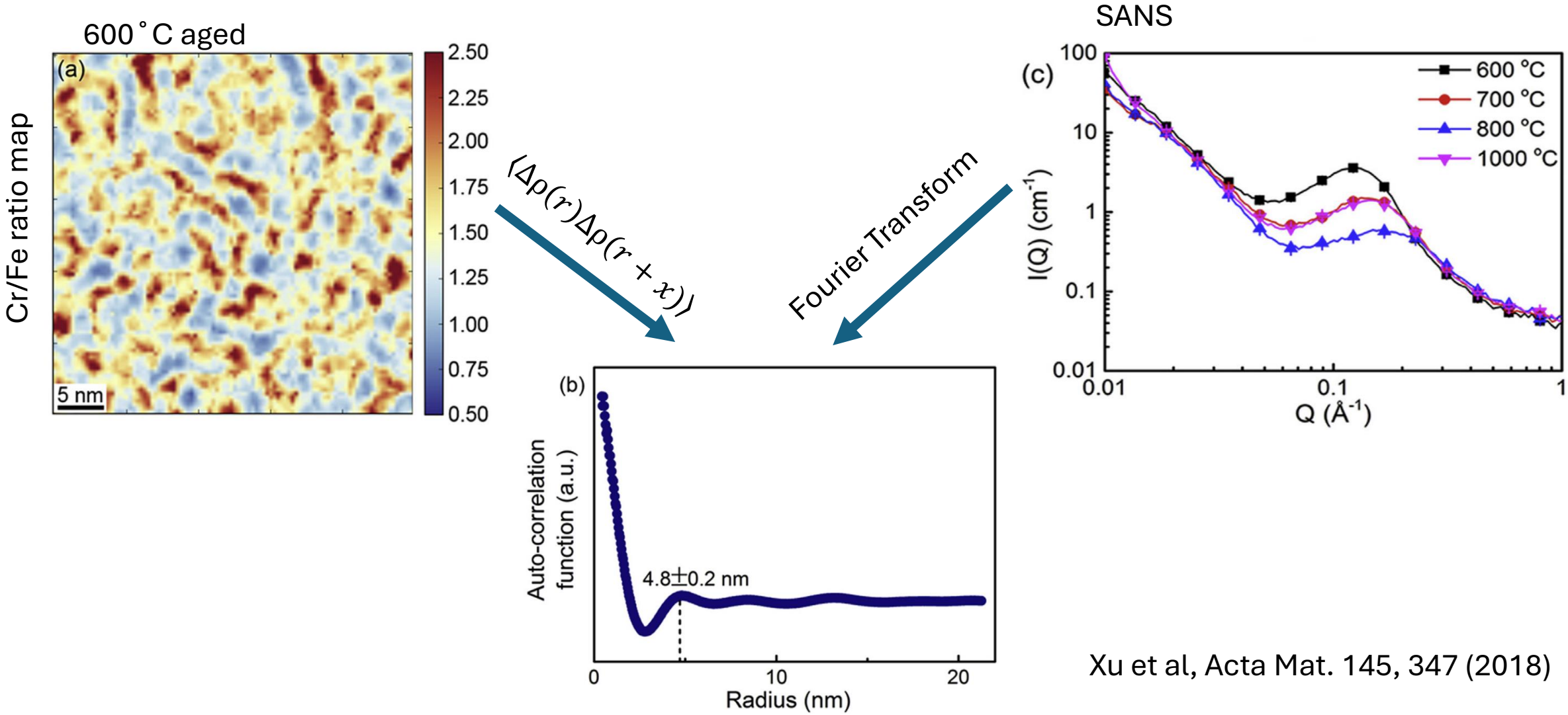
Science and Technology Facilities Council

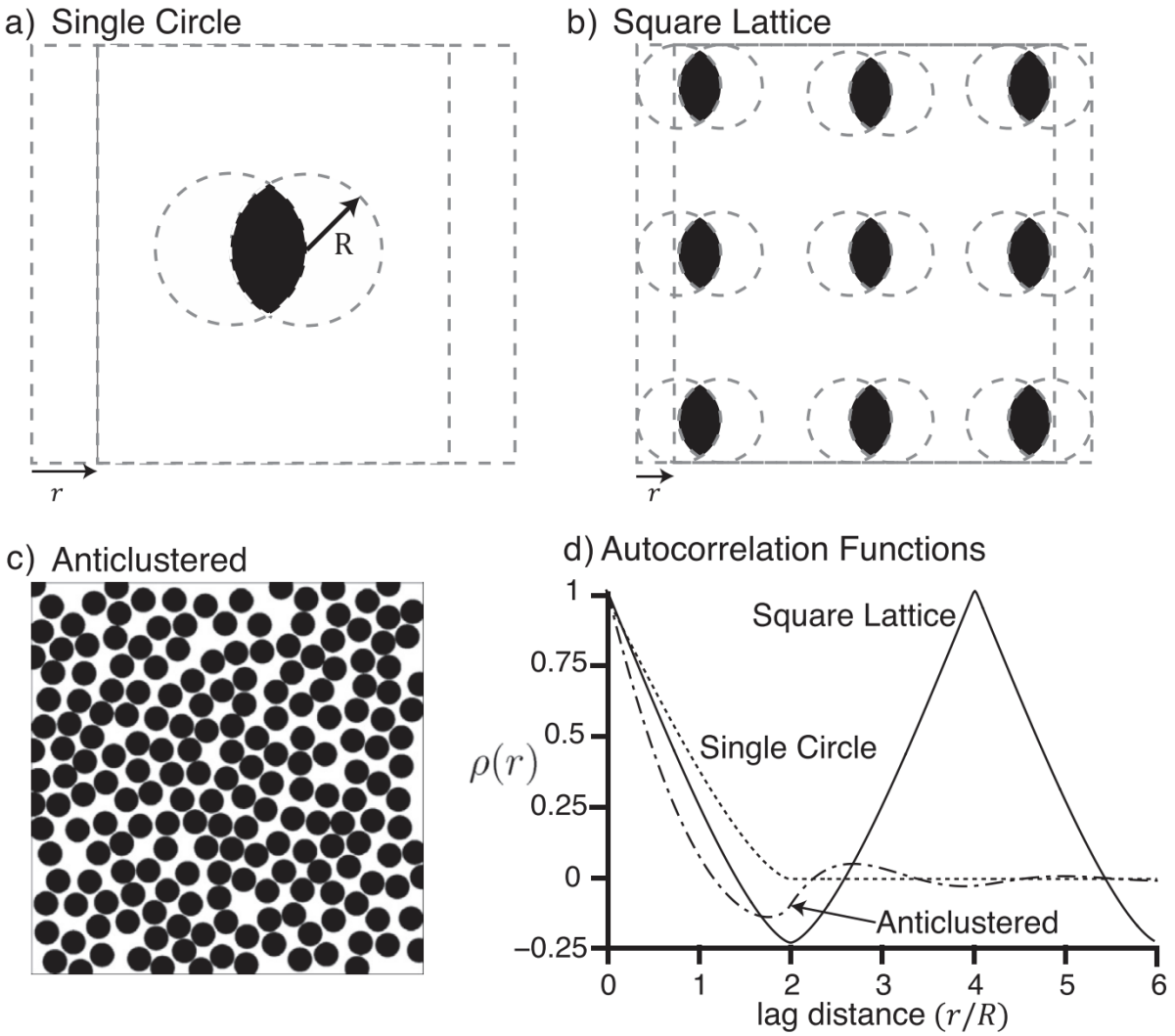
Correlation Functions in SasView

Example 1: Spinodal Decomposition in Fe-Cr binary alloys

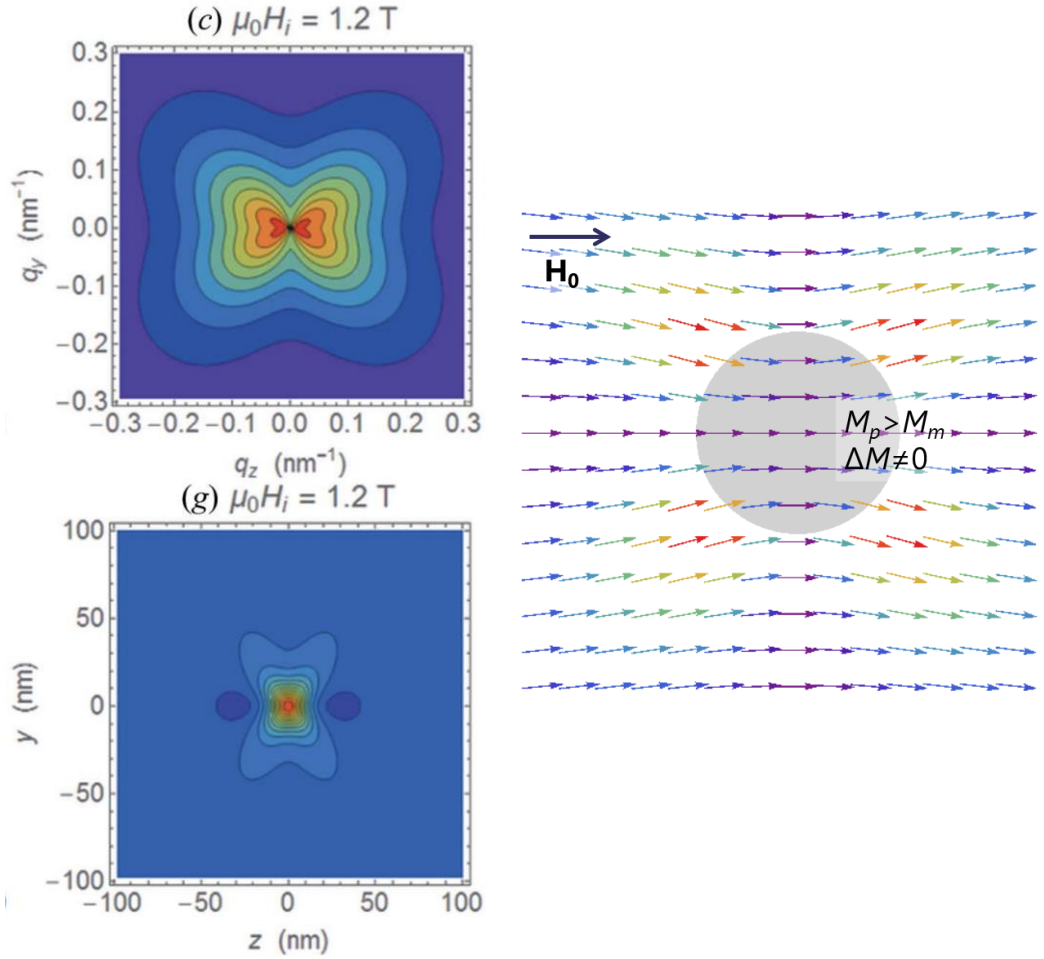


Xu et al, Acta Mat. 145, 347 (2018)

What measures the correlation function?

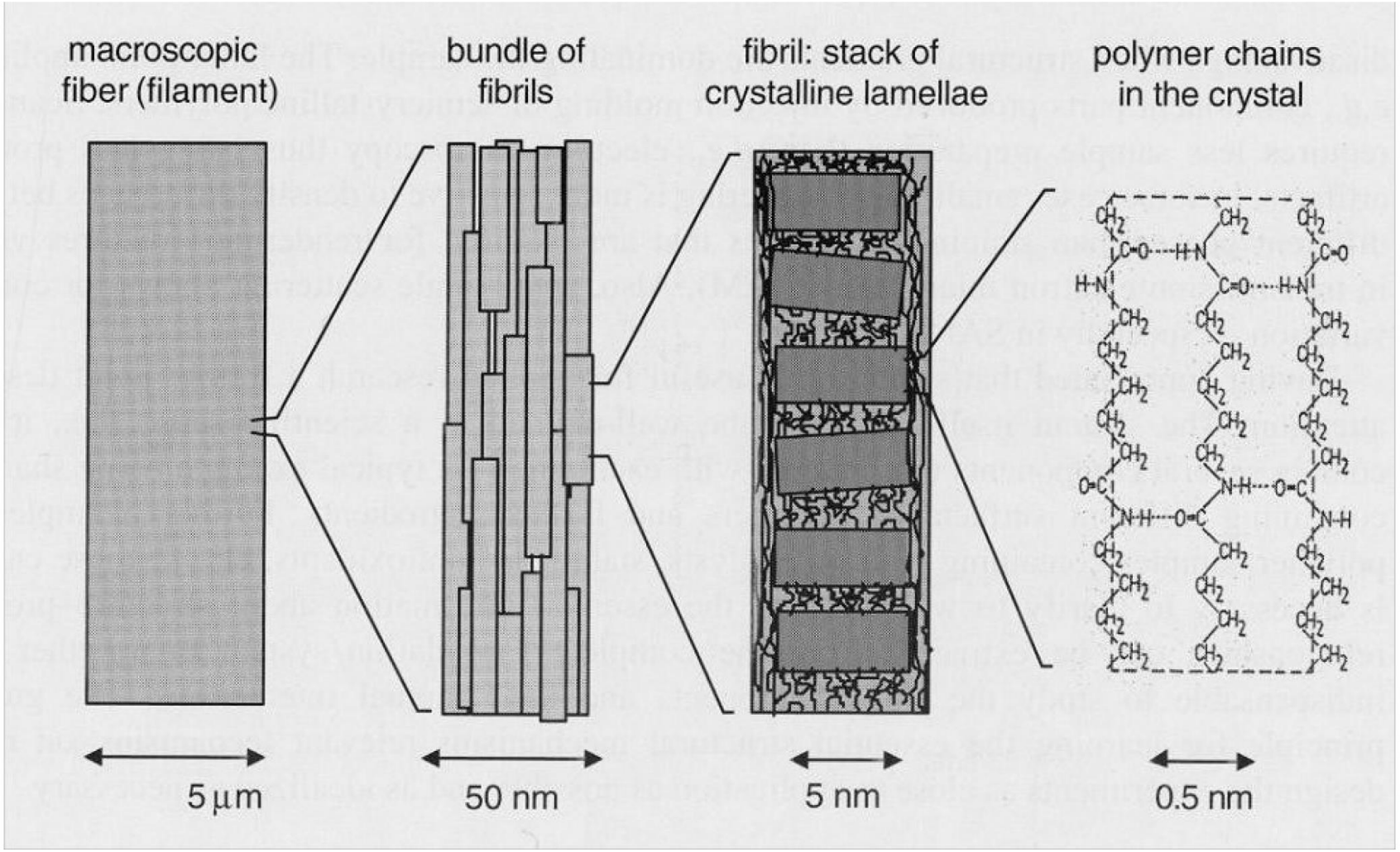


C. Thissen and M. Brandon, Journal of Structural Geology 81,135 (2015)



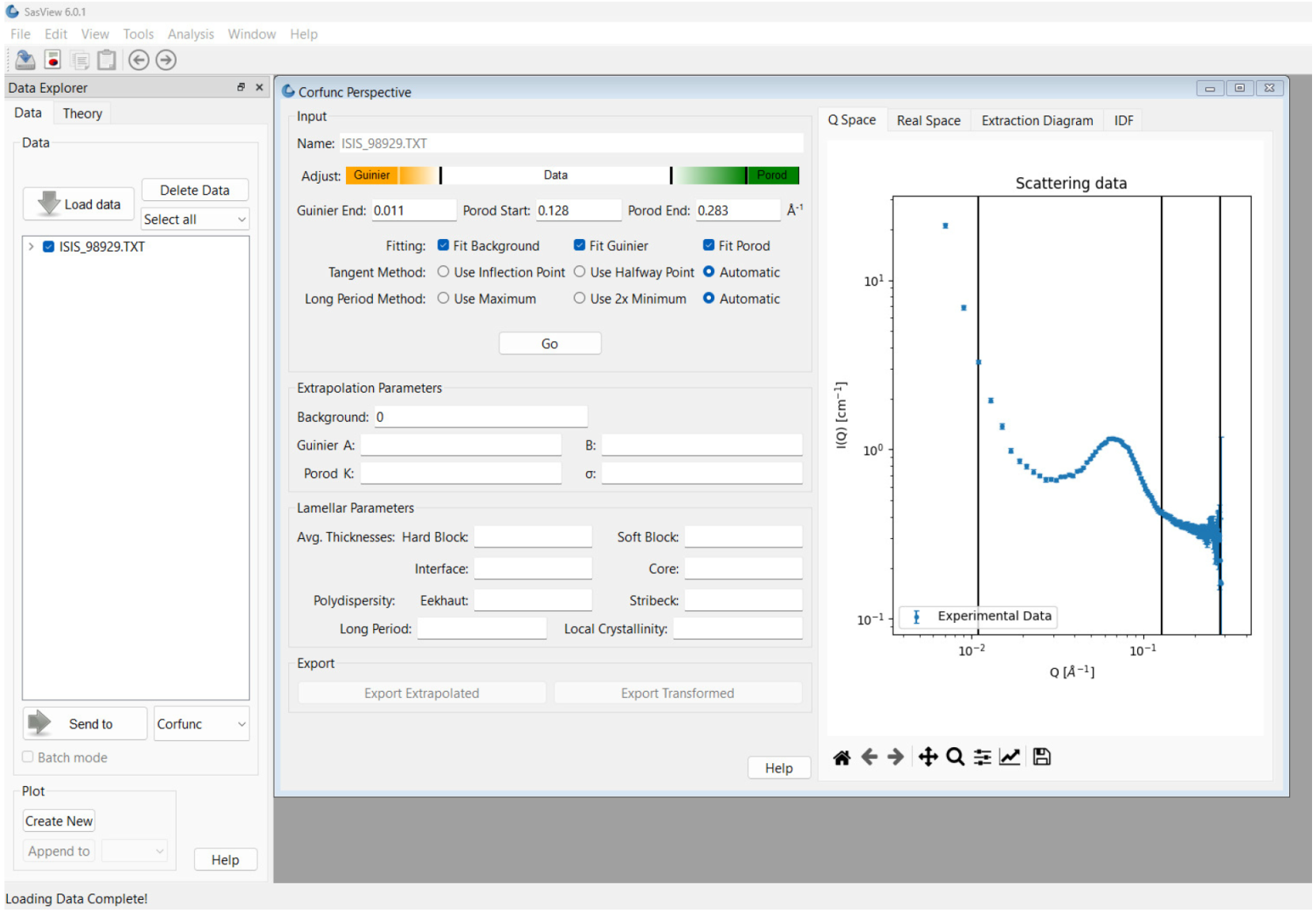
D.Mettus and A. Michels,J. Appl. Cryst. (2015). 48, 1437–1450

Example 2: Nylon as quasi-lamellar system



"Use of scattering methods in chemical industry - SAXS and SANS from fibers and films" (Chapter 21) of "Neutrons, X-rays and Light: Scattering Methods Applied to Soft Condensed Matter". Lindner & Zemb (editors), North-Holland (2002)

Example 2: Correlation function in Sasview



Example 2: Nylon as quasi-lamellar system

Tutorial
Let's go to Sasview

Example data under
\example_data\1d_data\ ISIS_98929.txt
in the SasView installation directory.

Checks on Correlation function results

Inspect $\Gamma_1(x)$ and $\Gamma_3(x)$ for artefacts arising from the extrapolation and transformation processes:

- Do they tend to 0 as x tends to ∞?
- Do they tend to 1 as x tends to 0?
- Do they smoothly curve onto the ordinate at x = 0?
- Are there any artefacts or ripples at x values corresponding to $2\pi / q_{max}$ or the two q values used for extrapolation?
- Do the features/peaks in the correlation functions actually correspond to the expected spacings in the sample?